Steering a swarm that owns a proof

Simulation 2 (asynchronous activation), setup `task003-symmetric`

LLM-free simulator; report of 16 October 2026; runs of 15 October 2026

Mirrors the Simulation 1 report of the same date: same sections, figures and equations. Same data as the 15 October version, reorga

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the 48 most recent posts on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved) silences it, posts the one fact from its pool that best supports its target, at most 30 messages per episode. Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true. The agents hold, together, a proof of A0; no single agent does, and neither pool can prove its target (Table 1).

Table 1: Facts available at $t = 0$ and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	18 distinct	(1, 0, 0): proves A0
A0 pool (controller with target A0)	12, of which 10 new to the agents	(0.963, 0.037, 0)
A2 pool (controller with target A2)	12, of which 10 new to the agents	(0.114, 0, 0.886)
agents' facts + either pool	28 distinct	(1, 0, 0)

The runs. Posts read by an agent $q \in \{3, 6, 12\}$; posts read by the controller $q_c \in \{3, 6, 12, 24\}$; controller rate $\lambda_c \in \{0.5, 1, 2, 3, 4, 5, 6, 7, 8\}$ (one fact per message); memory $\rho \in \{1, 0.75, 0.5, 0.25\}$; vote gate on and off; proof stop on; 1,000 episodes per setting (1,740 settings). Unless stated otherwise: $q = 6$, $q_c = 12$, $\lambda_c = 1$ (the main setting, which posts about as many facts as Simulation 1 with one fact per night), gate on. The vote gate is on throughout.

Summary. The 24 agents together own a proof of A0, and with full memory they keep it: no controller changes the swarm for more than a few time units. **Forgetting is what makes the swarm steerable**: the shorter the memory, the larger the effect, and the false controller wins the majority when $\rho \leq 0.5$ (Figure 1). Seen side by side (Figures 2, 4–6), **susceptibility and readability move together** (they rise, overshoot, persist and reverse together), while **the efficiency η hardly moves** (0.76–0.94): it measures how symmetric the two controllers' effects are, not how large. **The 30-message budget sets the end point and the rate sets the speed**: the slope of the susceptibility against the rate is large early and for the first unit of rate, and close to zero by $t = 30$ (Figures 7, 8); per fact, slow control is the most effective (Figure 3). **Reading more posts protects the proof when agents remember and helps the controller when they forget** (Figure 4). Every controller replaces part of the agents' own proofs (Figure 9).

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose last vote is answer a at time t and N_t the number of agents who have voted,

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

At $t = 0$ nobody has voted and we set $x_{a,0} = 0$; all agents have voted by $t \approx 5$. (The 9–14 October reports divided by 24, which counted agents who had not yet acted as voting against the target; this inflated the apparent rise of the shares and the apparent bias of the controller's sensing at early times.)

Paired susceptibility. Following BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\text{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G^{(0)}_t(\pi^+)$ for the controller targeting A0 and $G_t \equiv G^{(2)}_t(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\text{A0}, \text{A2}\}$ is drawn by a fair coin. The state X at time t is the vote of a randomly chosen agent, $X \in \{\text{A0}, \text{A1}, \text{A2}\}$. Each episode e of arm s (silent 0, target A0 $+$, target A2 $-$) contributes its vote shares $y^{(e)}_s(x) = x_{x,t}$, and

$$P_s(x) = \frac{1}{E}\sum_{e=1}^{E} y^{(e)}_s(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I(Z;X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z;X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z;X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004; without it, by at most 0.005 where C is finite. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted up to t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring rates,

$$\frac{\Delta G}{\Delta \lambda_c} = \frac{G(\lambda_c^{(k+1)}) - G(\lambda_c^{(k)})}{\lambda_c^{(k+1)} - \lambda_c^{(k)}}. \tag{8}$$

Episode e uses the same random draws at every rate, so the slope is estimated from per-episode differences, $[x^{(e)}_{\text{target},t}(\lambda_c^{(k+1)}) - x^{(e)}_{\text{target},t}(\lambda_c^{(k)})]/(\lambda_c^{(k+1)} - \lambda_c^{(k)})$, whose mean is Eq. (8) and whose standard error gives the 95% intervals (typically ± 0.01–0.03, about 30% narrower than for two independent means).

2 Votes over time

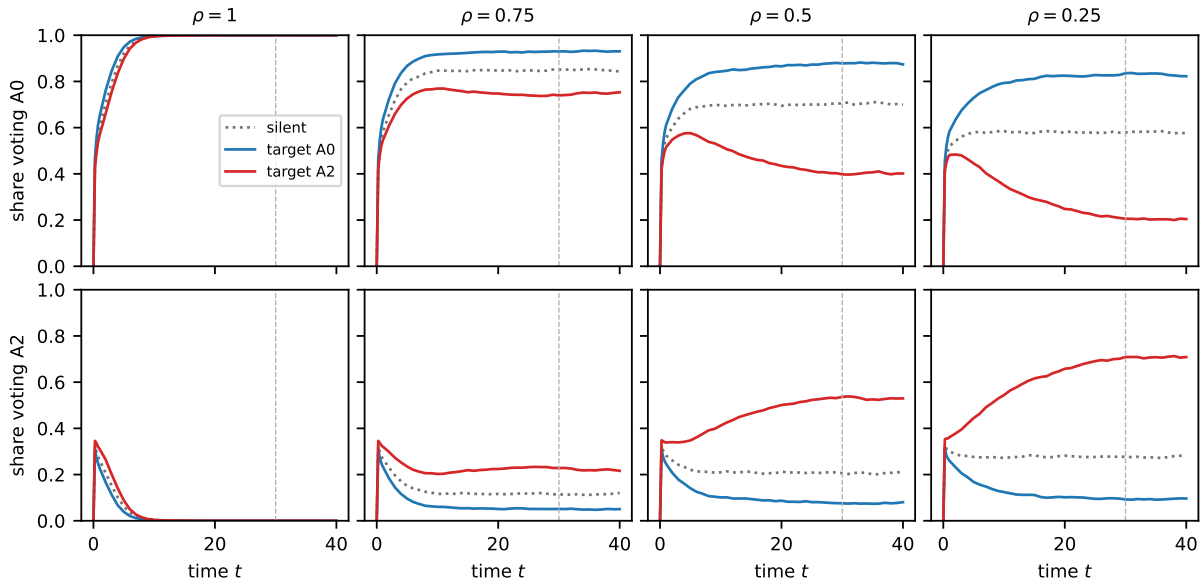


Figure 1: Share of agents voting A0 (top) and A2 (bottom), Eq. (1), without a controller (dotted), with the controller targeting A0 (blue) and A2 (red); one column per memory ρ. $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on. At $t = 0$ nobody has voted and every share is 0; after that the shares count only agents who have voted (all have by $t \approx 5$). The controller stops at $t = 30$ (dashed). 95% intervals are narrower than the lines (± 0.01).

- **The shorter the memory, the less firmly the silent swarm holds the truth**: 100%, 85%, 71% and 58% vote A0 at $t = 30$ for $\rho = 1, 0.75, 0.5, 0.25$.
- **With full memory no controller matters**: every agent votes A0 by $t \approx 10$ in all three arms.
- **The false controller wins the majority when** $\rho \leq 0.5$: A2 gets 54% ($\rho = 0.5$) and 71% ($\rho = 0.25$) of the votes, against 21% and 28% in silence; at $\rho = 0.75$ only 23%.
- **The truth controller restores the truth** to 84–93% at every $\rho < 1$.

3 Observables as time series

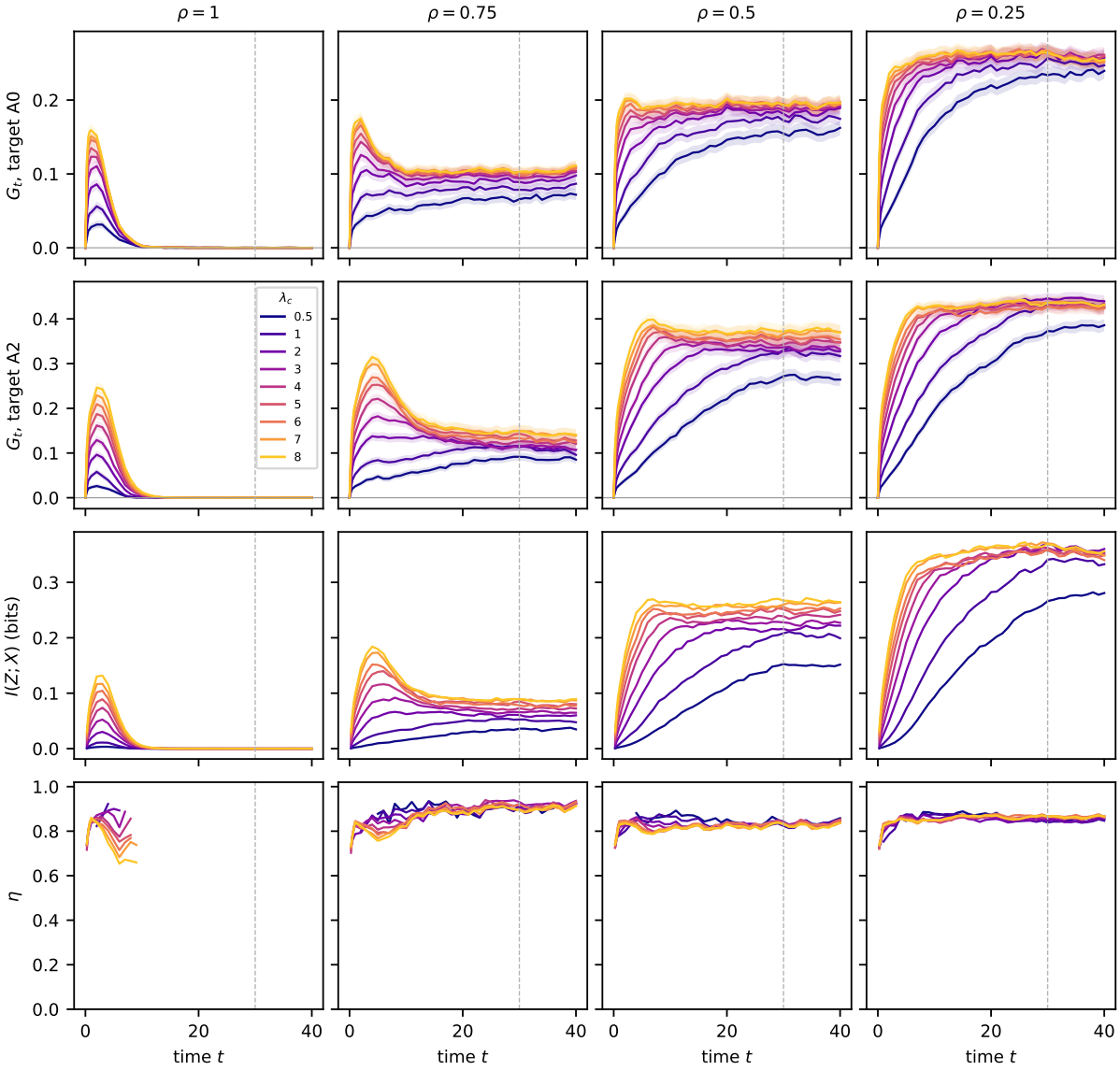


Figure 2: All observables over time, one column per memory ρ, one curve per controller rate λ_c; $q = 6$, $q_c = 12$, gate on. Rows: paired susceptibility G_t, Eq. (2), for target A0 and target A2 (bands: 95% intervals; $G_0 = 0$); readability $I(Z; X)$, Eq. (4); efficiency η, Eq. (6) (blank: undefined). The controller stops at $t = 30$ (dashed).

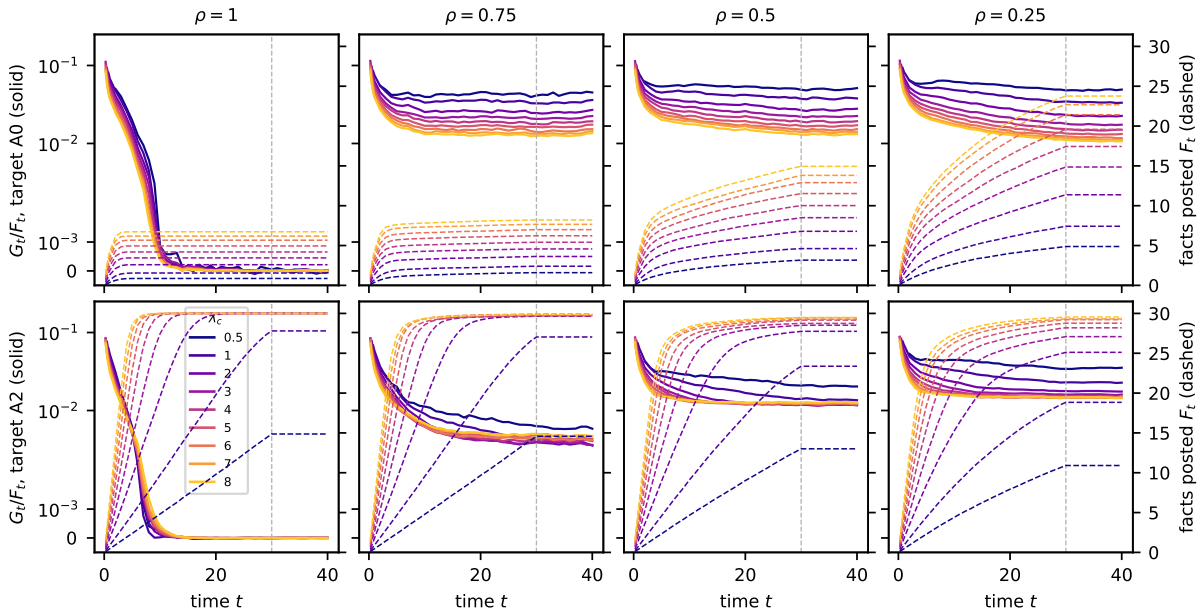


Figure 3: Gain per fact G_t/F_t (solid, left axis, log scale; shown once at least half a fact has been posted) and facts posted F_t (dashed, right axis), target A0 (top) and A2 (bottom), one colour per rate; $q = 6$, $q_c = 12$, gate on.

Susceptibility (Figure 2, rows 1–2).

- **It grows as memory shortens**: at $t = 30$, target A2, $G = 0.09$–0.15 ($\rho = 0.75$), 0.27–0.37 ($\rho = 0.5$), 0.37–0.44 ($\rho = 0.25$); target A0, 0.07–0.10, 0.16–0.19, 0.23–0.26.
- **With full memory it is only transient**: target A2 reaches 0.25 at $t \approx 2$ ($\lambda_c = 8$) and is gone by $t \approx 10$.
- **Fast controllers overshoot when agents remember well**: at $\rho = 0.75$, $\lambda_c = 8$, target A2 peaks at 0.32 ($t \approx 4$) and relaxes to 0.15; at $\rho \leq 0.5$ the peak is kept.
- **It persists after the controller stops**: from $t = 30$ to 40 it changes by at most 0.02.

Readability and efficiency (rows 3–4).

- **Readability has the shape of the susceptibility**: it rises, overshoots and persists where G does, and is largest at $\rho = 0.25$ ($I = 0.27$–0.37 bits at $t = 30$, against 0.04–0.09 at $\rho = 0.75$).
- **Efficiency does not**: after $t \approx 3$, η stays between 0.76 and 0.94 for every rate at $\rho < 1$, while G and I vary several-fold. Faster control raises the directed change and the total change together, so their ratio does not move; η reports the symmetry of the two controllers' effects, not their size.

Gain per fact and facts posted (Figure 3).

- **Each fact buys most at the start**: G_t/F_t is largest in the first time units and falls as facts accumulate.
- **The slowest controller is the most effective per fact**: at $t = 30$, target A2, $\rho = 0.5$: 0.021 per fact at $\lambda_c = 0.5$ (13 facts) against 0.013 at $\lambda_c = 8$ (30 facts); at $\rho = 0.25$: 0.034 against 0.015.
- **The false controller spends its budget, the truth controller does not**: target A2 reaches the 30-message budget by $t \approx 5$ at $\lambda_c = 8$; target A0 posts only 2–24 facts by $t = 30$ because the gate keeps it silent once the swarm agrees, which is why it gets more per fact (0.049 at $\rho = 0.5$, $\lambda_c = 0.5$).

4 Phase diagrams

All panels at $t = 30$, gate on; the susceptibility is for target A2 (grey number: its 95% interval includes 0); readability and efficiency are for the pair of targets. Colour scales are shared with the Simulation 1 report.

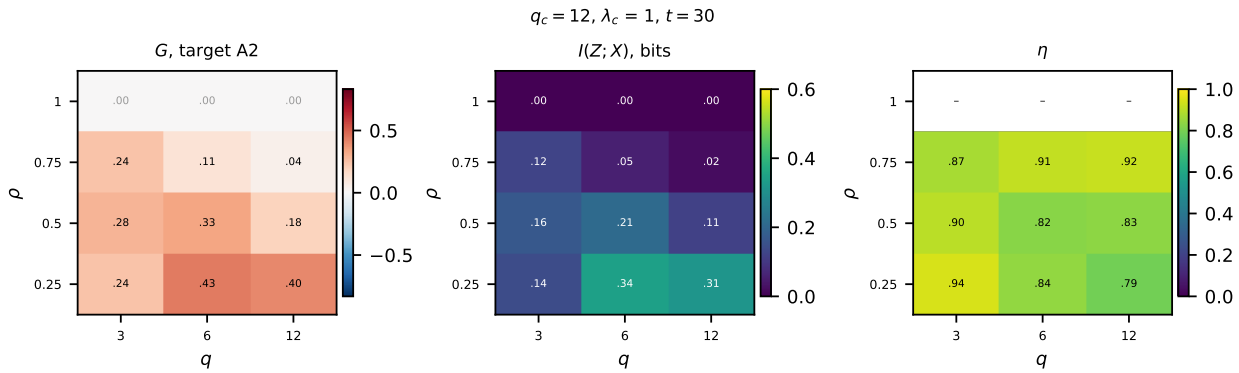


Figure 4: Susceptibility, readability and efficiency over memory ρ and the posts each agent reads q; $q_c = 12$, $\lambda_c = 1$.

- **Reading more defends the proof only when agents remember well**: at $\rho = 0.75$, G falls from 0.24 ($q = 3$) to 0.04 ($q = 12$) and I from 0.12 to 0.02 bits; at $\rho = 0.25$, G rises from 0.24 to 0.40–0.43 ($q = 6$, 12) and I from 0.14 to 0.31–0.34. A likely reason, not tested: with heavy forgetting the proof cannot be rebuilt anyway, and reading more only spreads the controller's facts faster.
- η **stays at 0.79–0.94** across this whole reversal.

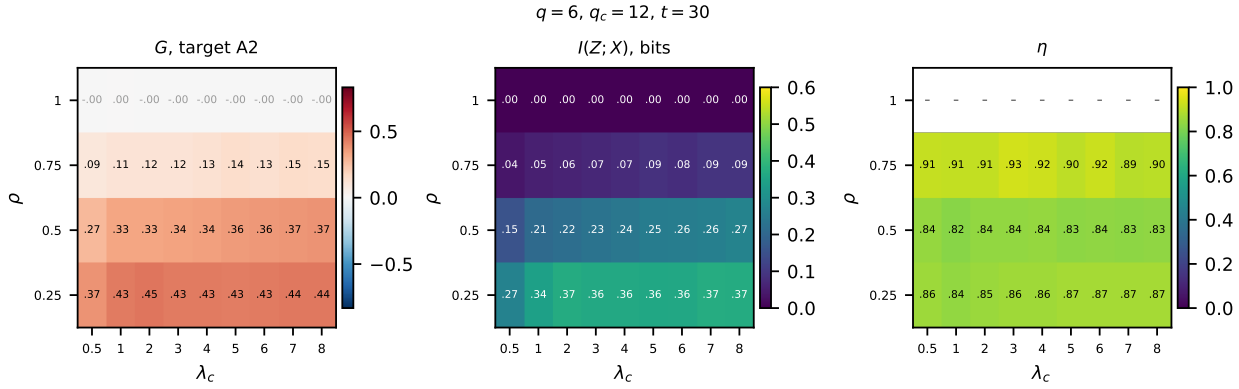


Figure 5: Susceptibility, readability and efficiency over memory ρ and the controller rate λ_c; $q = 6$, $q_c = 12$.

- **Memory sets the level, the rate barely changes it**: G goes from 0 at $\rho = 1$ to 0.37–0.45 at $\rho = 0.25$; along each row it changes by at most 0.10, almost all of it between $\lambda_c = 0.5$ and 1.
- G **and** I **move together** (rows ordered the same way; I from 0.04 to 0.37 bits); η **does not** (0.82–0.93, highest at $\rho = 0.75$ where the effects are smallest).

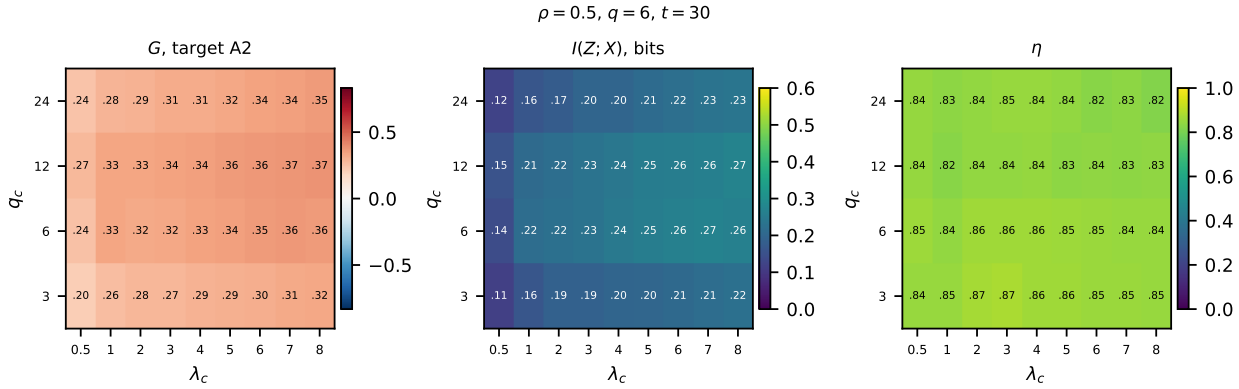


Figure 6: Susceptibility, readability and efficiency over the posts the controller reads q_c and its rate λ_c; $\rho = 0.5$, $q = 6$.

- **The controller's reading budget matters little**: mean G 0.28, 0.33, 0.34, 0.31 for $q_c = 3, 6, 12, 24$; I within 0.11–0.27 bits; η within 0.82–0.87.

- **Above $\lambda_c = 1$ little changes** along any row (G by at most 0.07), in any of the three observables: the budget is spent before $t = 30$.

5 Susceptibility against the rate, and its slope

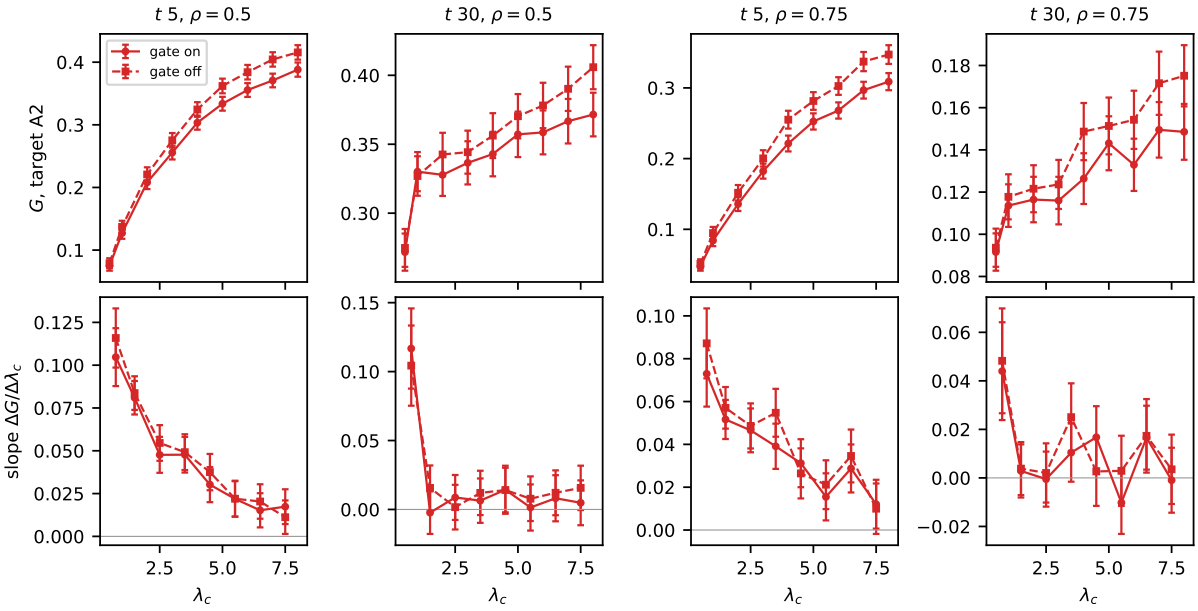


Figure 7: Top: G, target A2, against the controller rate λ_c, at $t = 5$ and $t = 30$ for $\rho = 0.5$ and 0.75; gate on (solid) and off (dashed); bars: 95% intervals. Bottom: the slope between neighbouring rates, Eq. (8), with paired 95% intervals.

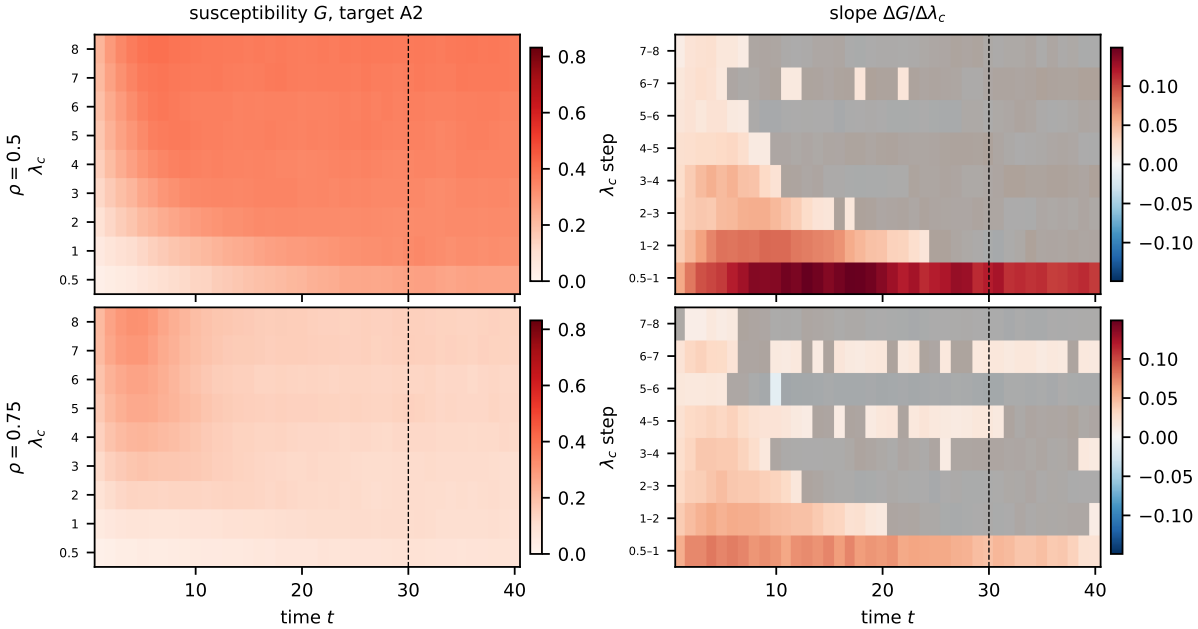


Figure 8: G, target A2 (left), and its slope between neighbouring rates (right) over time (x) and rate (y), for $\rho = 0.5$ (top) and 0.75 (bottom); $q = 6$, $q_c = 12$, gate on. Each row of the right panels is one step between neighbouring rates; faded cells: its paired 95% interval includes 0. The controller stops at $t = 30$ (dashed).

- **Early, speed buys susceptibility with diminishing returns** (Figure 7): at $t = 5$, $\rho = 0.5$, G rises from 0.08 ($\lambda_c = 0.5$) to 0.39 ($\lambda_c = 8$); the slope falls from 0.11 to about 0.02 per unit of rate.
- **The slope moves down the rate axis as time passes** (Figure 8, right): at $t = 5$, $\rho = 0.5$, every step is significantly positive (0.015–0.11); the steps above $\lambda_c = 1$ stop being distinguishable from 0 between

$t \approx 7$ and $t \approx 24$ (faded cells), and by $t = 30$ only the first step ($0.5 \to 1$, slope 0.12 ± 0.03) is.

- **Fast controllers get there early, slow ones late**: at $\rho = 0.5$, $\lambda_c = 8$ reaches 90% of its $t = 30$ value by $t \approx 3$, $\lambda_c = 0.5$ only by $t \approx 24$ (Figure 8, left).
- **The marginal susceptibility is the efficiency of adding rate**: high early and for the first unit of rate, close to zero once the budget is spent. Turning the gate off adds at most +0.03.

6 Who holds the proof

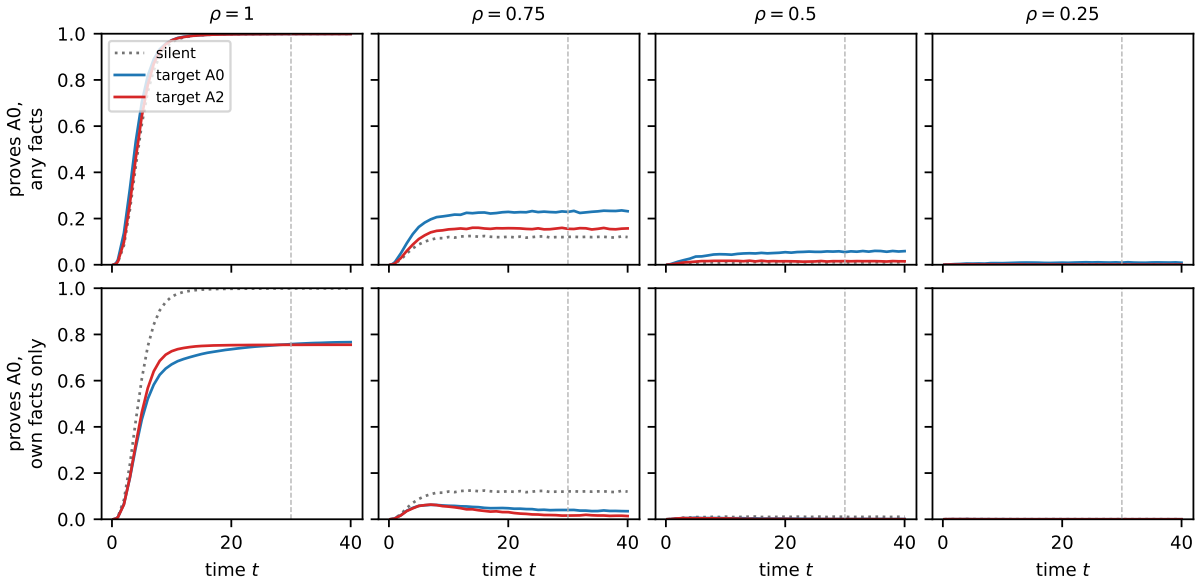


Figure 9: Share of agents whose remembered facts prove A0 (top) and prove it from the agents' original facts only (bottom); silent (dotted), target A0 (blue), target A2 (red); $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on.

- **Proofs need memory**: at $t = 30$ the silent swarm has 100%, 12%, 1% and 0% of agents proving A0 for $\rho = 1, 0.75, 0.5, 0.25$. Below $\rho = 0.75$ votes rest on beliefs, not proofs.
- **Every controller replaces the agents' own proofs**: with full memory all agents prove A0 under any arm, but only 76% from their own facts under either controller (100% in silence); at $\rho = 0.75$, 23% prove A0 under the truth controller but only 4% from their own facts.
- **The false controller's true facts complete proofs of A0** (16% against 12% in silence at $\rho = 0.75$).

7 Conclusions

1. **Memory decides whether a swarm that owns a proof can be steered.** With full memory every effect vanishes by $t \approx 10$. As memory shortens the silent swarm holds the truth less firmly (100%, 85%, 71%, 58% voting A0 at $t = 30$) and the susceptibility rises steadily.
2. **Susceptibility and readability move together; efficiency does not.** In every time series and every phase diagram, G and I rise, fall and reverse together, while η stays at 0.76–0.94: it measures how symmetric the two controllers' effects are, not how large.
3. **The budget sets the end point; the rate sets the speed.** The slope of G against the rate is large early and for the first unit of rate, and close to zero by $t = 30$; per fact, slow control is the most effective.
4. **The posts each agent reads switch sides with memory**: they defend the proof at $\rho = 0.75$ and help the controller at $\rho = 0.25$, in susceptibility and readability alike.
5. **Control replaces the agents' own proofs**, even when every agent ends up proving the truth.
6. **Compared with Simulation 1** (same settings, about the same facts posted): Simulation 2 is less susceptible at $\rho = 0.5$ (0.33 against 0.53, target A2) and more at $\rho = 0.25$ (0.43 against 0.31).

Caveats. One world, 24 exact-Bayesian agents voting by probability matching, one controller policy with a 30-message budget; 1,000 episodes per setting (95% intervals on G about ± 0.02). Information is plug-in from 1,000 episodes per arm. Data: `results/simulation_2/2026-10-15_sim2_symmetric_fine`; tables and figures: `analysis/task003_llm_free/shared_analysis/gold_standard/`.